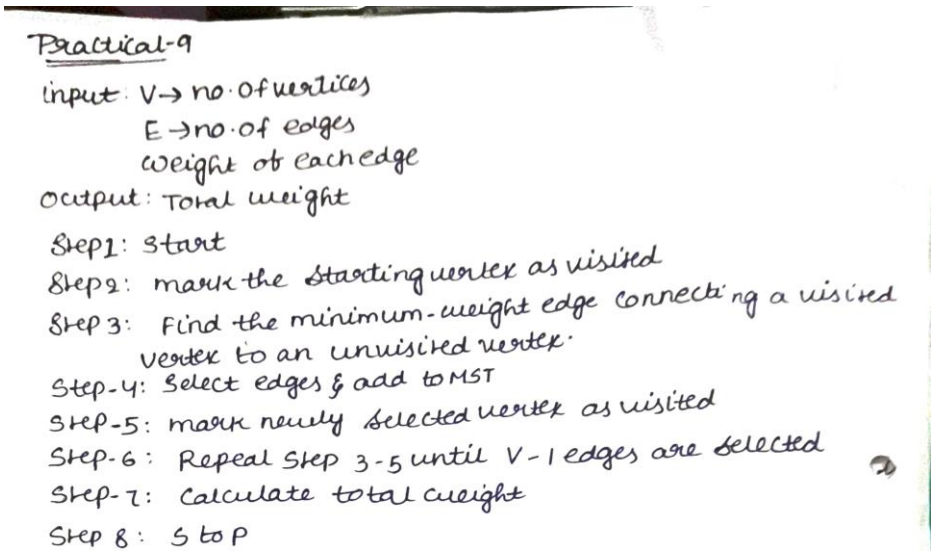
PRACTICAL-9

Aim: To Implementation of prim's algorithm.

Algorithm:


Practical-9

Input: V → no. of vertices
E → no. of edges
Weight of each edge

Output: Total weight

Step1: Start
Step2: mark the starting vertex as visited
Step 3: Find the minimum-weight edge connecting a visited vertex to an unvisited vertex.
Step-4: Select edges & add to MST
Step-5: mark newly selected vertex as visited
Step-6: Repeat Step 3-5 until V-1 edges are selected
Step-7: Calculate total weight
Step 8: Stop

Code:

```
#include <iostream>
using namespace std;

int main() {
   int V;
   int cost[100][100];
   int visited[100] = {0};

   cout << "Enter number of vertices: ";
   cin >> V;

   cout << "Enter adjacency matrix (0 if no edge):\n";

   for (int i = 0; i < V; i++) {
      for (int j = 0; j < V; j++) {
         cin >> cost[i][j];
      }
   }

   visited[0] = 1;

   int edges = 0;
```

```
    int minCost = 0;

    cout << "\nEdges in Minimum Spanning Tree:\n";

    while (edges < V - 1) {
        int min = 9999;
        int u = -1, v = -1;

        for (int i = 0; i < V; i++) {
            if (visited[i]) {
                for (int j = 0; j < V; j++) {
                    if (!visited[j] &&
                        cost[i][j] != 0 &&
                        cost[i][j] < min) {
                        min = cost[i][j];
                        u = i;
                        v = j;
                    }
                }
            }
        }

        if (u == -1) {
            cout << "MST is not possible. Graph is disconnected.\n";
            return 0;
        }

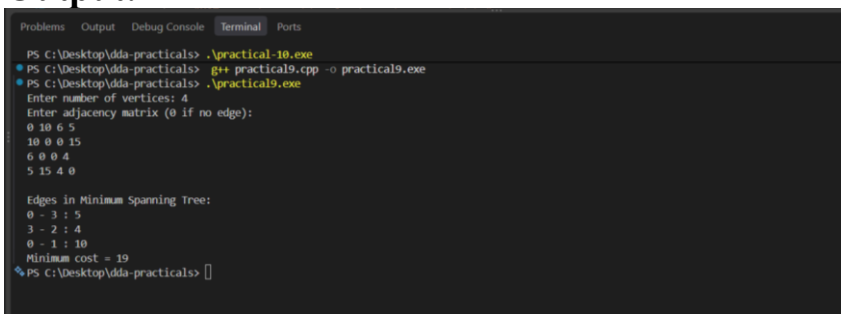
        cout << u << " - " << v
             << " : " << min << endl;

        visited[v] = 1;
        minCost += min;
        edges++;
    }

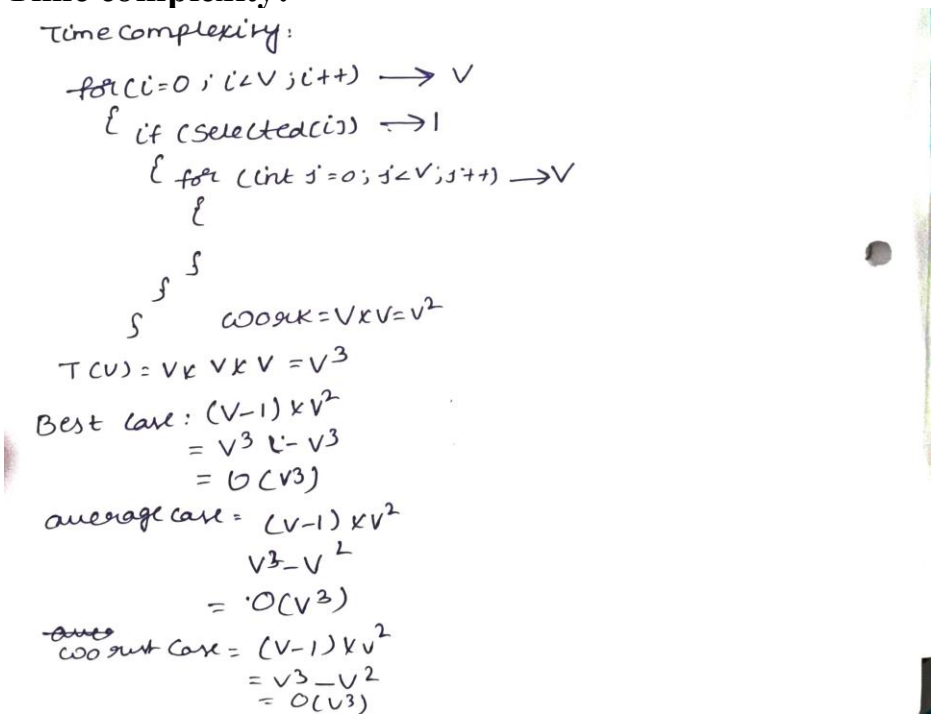
    cout << "Minimum cost = " << minCost << endl;

    return 0;
}
```

Output:



Time complexity:



Result:

Thus, Prim's algorithm was implemented successfully in C++ to find the Minimum Spanning Tree (MST) of a connected, weighted, undirected graph and calculate its minimum total cost. The time complexity of the implemented algorithm is (O(V^3)).